

Problem 1

- a. Minimize the cost function L on interval $a \leq x \leq b$ and verify by checking 2nd order optimality condition

$$L(k) = \frac{1}{2} \int_a^b [k - f(x)]^2 dx$$

Necessary condition:

$$\rightarrow \frac{\partial L}{\partial k} = \frac{1}{2} \int_a^b \frac{\partial}{\partial k} [k - f(x)]^2 dx = 0 \quad (\text{linear operator, Leibniz's rule})$$

$$\frac{\partial L}{\partial k} = \frac{1}{2} \int_a^b 2[k - f(x)] dx = 0$$

$$\frac{\partial L}{\partial k} = \left[(kx) \Big|_a^b - F(x) \Big|_a^b \right] = 0$$

$$\frac{\partial L}{\partial k} = k(b-a) - [F(b) - F(a)] = 0$$

Solving for k ,

$$k = \frac{F(b) - F(a)}{(b-a)}$$

Second order Optimality:

$$\frac{\partial^2 L}{\partial k^2} = \frac{\partial}{\partial k} \{ k(b-a) - [F(b) - F(a)] \} = \boxed{b-a}$$

Minimum iff $b > a$. (Sufficient condition)

- b. Determine optimal value of k and $L(k)$ for $f(x) = x^4$
 $a = 0$, $b = 1$.

$\rightarrow$ From a.:

$$k^* = \frac{F(b) - F(a)}{(b-a)} = \frac{\int_a^b f(x) dx}{(b-a)} = \frac{1/5}{1-0} = \boxed{1/5}$$

$$F(x) = \int_a^b f(x) dx = \int_0^1 x^4 dx = \frac{1}{5} x^5 \Big|_0^1 = \frac{1}{5}$$

$$\text{then } L(k^*) = \frac{1}{2} \int_0^1 [1/5 - x^4]^2 dx$$

$$= \frac{1}{2} \int_0^1 (1/25 - \frac{2}{5}x^4 + x^8) dx$$

$$= \frac{1}{2} \left[\left(\frac{1}{25}x \right) \Big|_0^1 + \left(-\frac{2}{25}x^5 \right) \Big|_0^1 + \left(\frac{1}{9}x^9 \right) \Big|_0^1 \right]$$

$$= \frac{1}{2} \left[\frac{1}{25} - \frac{2}{25} + \frac{1}{9} \right]$$

$$\boxed{L(k^*) = \frac{1}{50} - \frac{1}{25} + \frac{1}{18} = \frac{8}{225}}$$

1. Determine k^* , m^* , $Z(k^*, m^*)$:

$$Z(k, m) = \frac{1}{2} \int_a^b [mx + k - f(x)]^2 dx$$

Necessary condition:

$$\frac{\partial Z}{\partial \vec{p}} = \vec{0} = \left[\frac{\partial Z}{\partial k}, \frac{\partial Z}{\partial m} \right] = \left[\frac{1}{2} \int_a^b \frac{\partial}{\partial k} [mx + k - f(x)]^2 dx, \frac{1}{2} \int_a^b \frac{\partial}{\partial m} [mx + k - f(x)]^2 dx \right]$$

$$\frac{\partial Z}{\partial \vec{p}} = \vec{0} = \left[\frac{1}{2} \int_a^b 2(mx + k - f(x)) dx, \frac{1}{2} \int_a^b 2(mx + k - f(x))x dx \right]$$

for $f(x) = x^4$, $a = 0$, $b = 1$

$$\frac{\partial Z}{\partial \vec{p}} = \vec{0} = \left[\int_0^1 mx + k - x^4 dx, \int_0^1 mx^2 + kx - x^5 dx \right]$$

$$\frac{\partial Z}{\partial \vec{p}} = \vec{0} = \left[\left(\frac{1}{2} mx^2 \right)' \Big|_0^1 + (kx)' \Big|_0^1 + \left(-\frac{1}{5} x^5 \right)' \Big|_0^1, \left(\frac{mx^3}{3} \right)' \Big|_0^1 + \left(\frac{kx^2}{2} \right)' \Big|_0^1 + \left(-\frac{x^6}{6} \right)' \Big|_0^1 \right]$$

$$\frac{\partial Z}{\partial \vec{p}} = \vec{0} = \left[\frac{m}{2} + k - \frac{1}{5}, \frac{m}{3} + \frac{k}{2} - \frac{1}{6} \right]$$

Solving:

$$\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix} \begin{bmatrix} k^* \\ m^* \end{bmatrix} = \begin{bmatrix} 1/5 \\ 1/6 \end{bmatrix}$$

$$\begin{bmatrix} k^* \\ m^* \end{bmatrix} = \begin{bmatrix} -1/5 \\ 4/5 \end{bmatrix}$$

Sufficient condition:

$$\frac{\partial^2 Z}{\partial \vec{p}^2} = \begin{bmatrix} \frac{\partial^2 Z}{\partial k^2} & \frac{\partial^2 Z}{\partial k \partial m} \\ \frac{\partial^2 Z}{\partial m \partial k} & \frac{\partial^2 Z}{\partial m^2} \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}$$

Constant Hessian, thus:

$$\det(A - \lambda I_m) = 0$$

$$\det \begin{bmatrix} 1-\lambda & 1/2 \\ 1/2 & 1/3-\lambda \end{bmatrix} = 0$$

$$(1-\lambda)(1/3-\lambda) - 1/4 = 0$$

$$\lambda^2 - 4/3\lambda + 1/12 = 0$$

$$\lambda = \frac{4/3 \pm \sqrt{(4/3)^2 - 4(1/12)}}{2(1)} = \frac{2}{3} \pm \frac{\sqrt{13}}{6} > 0; \text{ both positive \& thus positive definite}$$

$\therefore$ minimum

$$\begin{aligned}
 Z(4^{\text{th}}, 20^{\text{th}}) &= \frac{1}{2} \int_0^1 \left[\frac{4}{3}x + \frac{1}{3} - x^4 \right]^2 dx \\
 &= \frac{1}{2} \int_0^1 x^8 - \frac{8x^5}{3} + \frac{2x^4}{3} + \frac{16x^2}{9} - \frac{8x}{3} + \frac{1}{9} dx \\
 &= \frac{1}{2} \left[\frac{x^9}{9} - \frac{8x^6}{18} + \frac{2x^5}{15} + \frac{16x^3}{27} - \frac{8x^2}{6} + \frac{x}{9} \right]_0^1 \\
 &= \frac{1}{2} \left[\frac{1}{9} - \frac{4}{9} + \frac{2}{15} + \frac{16}{27} - \frac{4}{3} + \frac{1}{9} \right] \\
 &= \frac{1}{2} \left[\frac{4}{225} \right] \\
 &= \boxed{\frac{2}{225}}
 \end{aligned}$$

Problem 2

Find min distance from $(0, 2)$ to curve $y = x^2 + 3x - 6$.
 Use a Lagrange multiplier to find value for x , solve for x^* , y^* point. Plot in MATLAB solution. What is optimal distance?

→ Distance formula:

$$d = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2}$$

for the point $(0, 2)$

$$d = \sqrt{(x)^2 + (y - 2)^2}$$

"Pseudo Hamiltonian"

$$Z = x^2 + (y - 2)^2$$

- Constant Hamiltonian with Lagrange multipliers

$$H = L + \lambda^T f$$

$$H = x^2 + (y - 2)^2 + \lambda_1 (x^2 + 3x - 6 + y)$$

1 constraint → 1 Lagrange multiplier.

2 variables, 1 constraint, 1 control, → $\bar{u} = u = y$
 Necessary conditions:

$$\text{I } \frac{\partial H}{\partial x} = 0 = 2x + 2x\lambda_1 + 3\lambda_1 = 2x + \lambda_1(2x + 3) = 0$$

$$\text{II } \frac{\partial H}{\partial u} = \frac{\partial H}{\partial y} = 0 = 2(y - 2) - \lambda_1 = 0$$

$$\text{III } \frac{\partial H}{\partial \lambda} = x^2 + 3x - 6 + y = 0$$

3 equations (non linear) 3 unknowns.

$$\text{III. } y = x^2 + 3x - 6$$

$$\text{III} \rightarrow \text{II. } \lambda_1 = 2(y - 2) = 2(x^2 + 3x - 8) = 2x^2 + 6x - 16$$

$$\rightarrow \text{I. } 2x + \lambda_1(2x + 3) = 0$$

$$\rightarrow 2x + (2x^2 + 6x - 16)(2x + 3) = 0$$

$$2x + 4x^3 + 12x^2 - 32x + 6x^2 + 18x - 48 = 0$$

$$4x^3 + 18x^2 - 12x - 48 = 0$$

$$\boxed{2x^3 + 9x^2 - 6x - 24 = 0} \quad \text{cubic eqn.}$$

Solved for roots with MATLAB.

SEE ATTACHED LIVESCREEN

$$\bar{x}^* = \begin{bmatrix} -4.5833 \\ 1.6603 \\ -1.5770 \end{bmatrix}$$

$$\rightarrow \text{II} \rightarrow \bar{\lambda}^* = \begin{bmatrix} -1.4865 \\ -0.5254 \\ -20.4882 \end{bmatrix}$$

$$\rightarrow \text{I} \rightarrow \bar{y}^* = \begin{bmatrix} 1.2568 \\ 1.7373 \\ -8.2441 \end{bmatrix}$$

$$\rightarrow \text{distance} \rightarrow \bar{d}^* = \begin{bmatrix} 4.6432 \\ 1.6809 \\ 10.3647 \end{bmatrix} \rightarrow \boxed{\text{optimal distance is } 1.6809}$$

Problem 3

Find the point nearest $(0,0,0)$ via Lagrange multipliers on the line defined by the intersection of

$$3x + 2y + z = 1 \quad \& \quad x + 2y - 3z = 4$$

Confirm by checking 2nd order optimality. What is the relationship between this line and the line from the intersection to the origin?

$$\rightarrow d = \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$$

$$\text{to origin, } x_0 = y_0 = z_0 = 0$$

$$d = \sqrt{x^2 + y^2 + z^2}$$

primal function:

$$L = x^2 + y^2 + z^2; \quad \bar{f} = [3x + 2y + z - 1, x + 2y - 3z - 4]$$

Hamiltonian with two constraints:

$$H = L + \bar{\lambda}^T \bar{f}$$

$$H = x^2 + y^2 + z^2 + \lambda_1 (3x + 2y + z - 1) + \lambda_2 (x + 2y - 3z - 4)$$

$$\bar{x} = [x, y] \quad u = z$$

$$H_x = [2x + 3\lambda_1 + \lambda_2, 2y + 2\lambda_1 + 2\lambda_2]$$

$$H_u = 2z + \lambda_1 - 3\lambda_2$$

$$H_\lambda = [3x + 2y + z - 1, x + 2y - 3z - 4]$$

$$\text{Solve: } \begin{bmatrix} 2 & 0 & 0 & 3 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 2 & 1 & -3 \\ 3 & 2 & 1 & 0 & 0 \\ 1 & 2 & -3 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} x^* \\ y^* \\ z^* \\ \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 0.2556 \\ 0.5556 \\ -0.8778 \\ 0.0222 \\ -0.5778 \end{bmatrix}$$

Second Order Optimality

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} \bar{z}^T & I_m \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} \bar{z} \\ I_m \end{bmatrix}$$

$$\bar{z} = -(\bar{f}_x)^T \bar{f}_u = -\left(\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \right)^{-1} \cdot \begin{bmatrix} 1 \\ -3 \end{bmatrix} = -\begin{bmatrix} 1/2 & -1/2 \\ -1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ -5/2 \end{bmatrix}$$

$$\bar{f}_x = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \quad \bar{f}_u = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

→

$m \times (n+m)$ $(n+m \times n+m)$ $(m+m \times m)$

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} 2 & -5/2 & 1 \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xy} \\ H_{yx} & H_{yy} \end{bmatrix} \begin{bmatrix} 2 \\ -5/2 \\ 1 \end{bmatrix} = 45/2$$

$$H_x = [2x + 3d_1 + d_2, 2y + 2d_1 + 2d_2]$$

$$H_{xx} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$H_{xy} = [2d_1 + d_2, -3d_2]$$

$$H_{yy} = 2$$

$$H_{yx} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$H_{yy} = [0 \ 0]$$

$$\frac{\partial^2 L}{\partial \bar{u}^2} = 45/2 > 0 \text{ is } (1) \rightarrow 45/2$$

positive $\rightarrow$ definite positive
 $\therefore$ is a minimum!

The relationship between the intersecting line and the line to the origin is that these lines are perpendicular to each other.

To check, the unit vector in the direction of the intersecting line and the unit vector in the direction of the line connecting to the origin will be orthogonal to each other, i.e.,

$$\hat{r}_1 \cdot \hat{r}_2 = 0 \rightarrow v_1 v_2 \cos \theta = 0 \rightarrow \theta = 90^\circ$$

$\therefore$ orthogonal.

Problem 2

Determine the stationary points by solving for the roots:

```
clc
clf

x = roots([2 9 -6 -24])
```

```
x = 3×1
    -4.5833
     1.6603
    -1.5770
```

```
lambda = (2.*x.^2)+(6.*x)-16
```

```
lambda = 3×1
    -1.4865
    -0.5254
   -20.4882
```

```
y = x.^2+3.*x-6
```

```
y = 3×1
     1.2568
     1.7373
    -8.2441
```

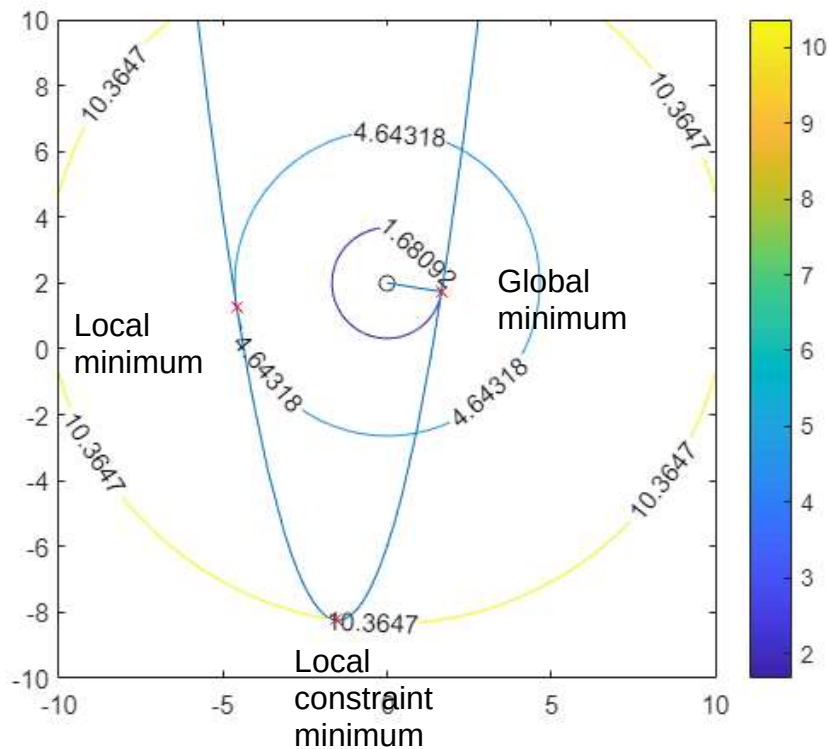
Taking the stationary points and determining optimal distance:

```
distance = sqrt((x.^2)+(y-2).^2)
```

```
distance = 3×1
     4.6432
     1.6809
    10.3647
```

Plot of solution and constraint

```
[u1, u2] = meshgrid(-10:0.1:10, -10:0.1:10);
d = sqrt(u1.^2+(u2-2).^2);
[C,h] = contour(u1,u2,d, [distance(1) distance(2) distance(3)]);
colorbar
set(h, "ShowText", "on")
hold on
fplot(@(u) u.^2+3*u-6)
axis equal
axis([-10 10 -10 10])
plot(0,2,'ko')
plot(x(1),y(1),"rx")
plot(x(2),y(2),"rx")
plot(x(3),y(3),"rx")
line([0,x(2)],[2,y(2)])
```



Problem 3

Stationary points found by solving system of linear equations generated from necessary condition (first order optimality) of the Hamiltonian:

```
answer = [2 0 0 3 1; 0 2 0 2 2; 0 0 2 1 -3; 3 2 1 0 0; 1 2 -3 0 0]^(-1)*[0;0;0;1;4]
```

```
answer = 5x1
    0.2556
    0.5556
   -0.8778
    0.0222
   -0.5778
```

Second order optimality:

```
matrix = [2 1 0; -5/2 0 1]*[2 0 0; 0 2 0; 0 0 2]*[2 -5/2; 1 0; 0 1]
```

```
matrix = 2x2
    10.0000   -10.0000
   -10.0000    14.5000
```

Determination of concavity:

```
eigs(matrix)
```

```
ans = 2x1
    22.5000
     2.0000
```

All eigen values positive, thus positive definite and solution is a minimum.

The associated optimal distance:

```
sqrt(answer(1)^2+answer(2)^2+answer(3)^2)
```



```
ans = 1.0698
```

Check solution with fmincon:

```
[x_opt, fval] = fmincon(@(x) x(1)^2+x(2)^2+x(3)^2, [0, 0, 0], [], [], [], [], [], [], @myConstraints)
```

Local minimum found that satisfies the constraints.

Optimization completed because the objective function is non-decreasing in feasible directions, to within the value of the optimality tolerance, and constraints are satisfied to within the value of the constraint tolerance.

<stopping criteria details>

```
x_opt = 1×3
```

```
    0.2556    0.5556   -0.8778
```

```
fval = 1.1444
```

Agreement in stationary points for x^* , y^* , and z^* .

Plot of Problem 3 solution and line to origin.

```
a1 = 3;
b1 = 2;
c1 = 1;
d1 = -1;
a2 = 1;
b2 = 2;
c2 = -3;
d2 = -4;

[x, y] = meshgrid(-10:1:10, -10:1:10);

z1 = (-a1 * x - b1 * y - d1) / c1;
z2 = (-a2 * x - b2 * y - d2) / c2;

figure;
hold on;
surf(x, y, z1, 'FaceColor', 'r', 'FaceAlpha', 0.5);
surf(x, y, z2, 'FaceColor', 'b', 'FaceAlpha', 0.5);

plot3(0,0,0,'kx')

axis equal
axis([-10 10 -10 10 -10 10])

n1 = [a1, b1, c1];
n2 = [a2, b2, c2];
line_dir = cross(n1, n2);

A = [a1, b1, c1; a2, b2, c2];
B = [-d1; -d2];
point_on_line = A \ B;

t = linspace(-10, 10, 100);
line_x = point_on_line(1) + line_dir(1) * t;
line_y = point_on_line(2) + line_dir(2) * t;
line_z = point_on_line(3) + line_dir(3) * t;
plot3(line_x, line_y, line_z, 'k', 'LineWidth', 2);

plot3([0, point_on_line(1)], [0, point_on_line(2)], [0, point_on_line(3)], 'g', 'LineWidth', 2);

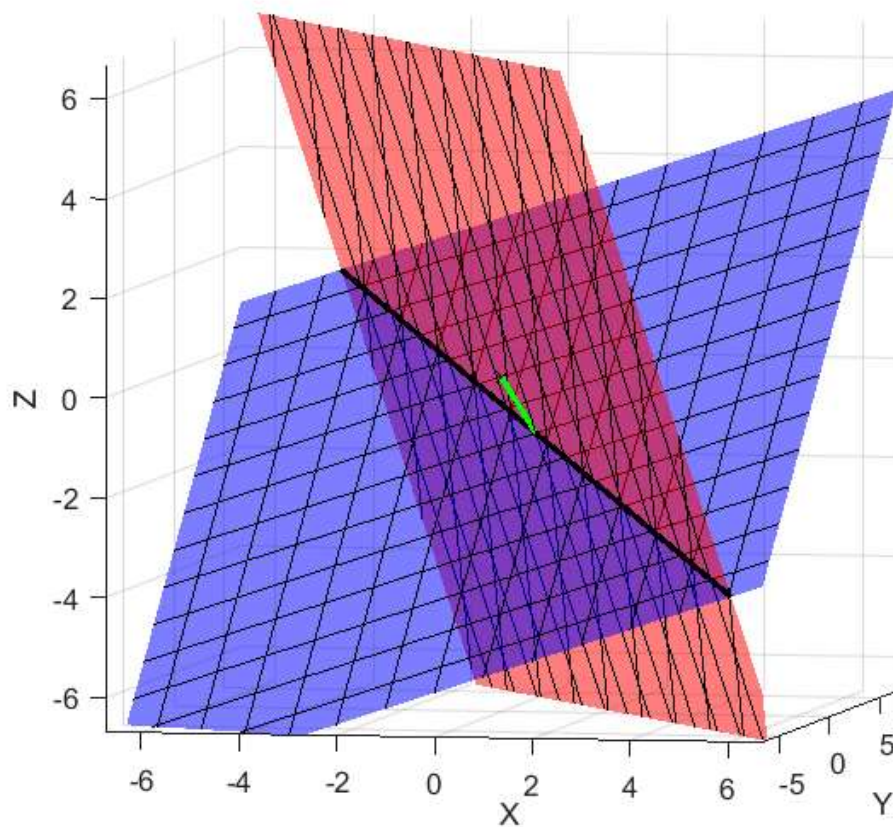
xlabel('X');
ylabel('Y');
zlabel('Z');
```

```

grid on;
axis equal;
hold off;

xlim([-6.7 6.7])
ylim([-6.7 6.7])
zlim([-6.7 6.7])
view([11.6 4.5])

```



```

function [c,ceq] = myConstraints(x)
    a1 = 3;
    b1 = 2;
    c1 = 1;
    d1 = -1;
    a2 = 1;
    b2 = 2;
    c2 = -3;
    d2 = -4;
    ceq(1) = a1 * x(1) + b1 * x(2) + c1 * x(3) + d1;
    ceq(2) = a2 * x(1) + b2 * x(2) + c2 * x(3) + d2;

    c = [];
end

```